Zeki Bora ÖN (PhD)

http://www.mu.edu.tr/en/personel/boraon
Email: boraon@mu.edu.tr & zekiboraon@gmail.com
Mobile: +90 552 305 3631
https://zboraon.github.io/

Muğla SK Üniversitesi
Mühendislik Fakültesi
Elektrik Elektronik Mühendisliği Bölümü
48000, Kötekli/Muğla/Türkiye

Education

- **İstanbul Teknik Üniversitesi (İTÜ)** — İstanbul, Türkiye
  Ph.D., Earth Science, under M. Sinan Özeren's supervision — *November 2018*
- **Muğla Üniversitesi** — Muğla, Türkiye
  M.S., Geological Engineering — *June 2013*
- **Boğaziçi Üniversitesi** — İstanbul, Türkiye
  B.S., Mathematics — *June 2009*

Academic Experience

- **Muğla SK Üniversitesi** — Muğla, Türkiye
  Lecturer at Electrical & Electronics Engineering Department — *August 2025 – Present*
  Lecturer at Geological Engineering Department — *January 2019 – July 2025*

  As a lecturer I taught the following undergraduate and graduate* courses:
  - Statistics and probability
  - Introduction to programming
  - Computer applications in geological engineering
  - Discrete mathematics
  - Signals and systems
  - Paleoclimatology*
- **Muğla SK Üniversitesi** — Muğla, Türkiye
  Research Assistant — *February 2015 – December 2018*

  As a research assistant I taught / assisted the following undergraduate and graduate* courses:
  - Statistics and probability
  - Introduction to programming
  - Historical geology and paleontology
  - Marine geology
  - Field geology
  - Introduction to paleoclimate
  - Quaternary environmental change*
  - Paleoclimatology*
  - Holocene climate variability at western Anatolia, comparison with Middle East and Europe*
  - Paleoceanography*
- **İTÜ** — İstanbul, Türkiye
  Research Assistant — *September 2013 – February 2015*
- **Muğla SK Üniversitesi** — Muğla, Türkiye
  Research Assistant — *February 2012 – September 2013*

Research Interests

- Climatology & Paleoclimatology
- Statistical methods
- Applications of information theoretical methods
- Stochastic processes
- Bayesian statistics

Papers

- Z. B. Ön. Bayesian stochastic imputation of discontinuous paleoenvironmental proxy records: a case study from Lake Ohrid stable isotopes. *Environmental and Ecological Statistics*, 33(1):129–148, 2026b. doi:10.1007/s10651-025-00701-6
- H. Çaldırak, Z. B. Ön, M. S. Akkiraz, M. N. Çağatay, B. Ghaleb, S. Wulf, K. K. Eriş, D. Acar, J. Kaiser, and S. Akçer-Ön. Chronology and environmental changes from a sediment core spanning the last 485 ka from Lake Acıgöl (SW Anatolia). *Quaternary International*, 722:109710, 2025a. doi:10.1016/j.quaint.2025.109710
- Z. B. Ön, M. E. Ateş, and J. Kaiser. A conceptual and statistical framework for delineating the timing of a stratigraphic transition: Holocene–Anthropocene boundary as a case study. *Progress in Physical Geography: Earth and Environment*, 49(1-2):63–83, 2025. doi:10.1177/03091333241310742
- I. B. Danladi, S. Akçer-Ön, T. Litt, Z. B. Ön, and L. Wacker. A late Holocene climate reconstruction from the high-altitude Lake Gölcük sedimentary records, Isparta (SW Anatolia). *Quaternary Research*, 115:120–133, 2023b. doi:10.1017/qua.2023.26
- Z. B. Ön. Dates of Holocene environmental changes in Lake Bafa: A hierarchical Bayesian analysis of change points. *Turkish Geographical Review*, 82:23–36, 2023d. doi:10.17211/tcd.1283443
- Z. B. Ön. A Bayesian change point analysis re-examines the 4.2 ka BP event in southeast Europe and southwest Asia. *Quaternary Science Reviews*, 312:108163, 2023e. doi:10.1016/j.quascirev.2023.108163
- Z. B. Ön, N. Macdonald, S. Akçer-Ön, and A. M. Greaves. A novel Bayesian multilevel regression approach to the reconstruction of an eastern Mediterranean temperature record for the last 10,000 years. *The Holocene*, 33(7):807–815, 2023b. doi:10.1177/09596836231163508
- I. B. Danladi, S. Akçer-Ön, Z. B. Ön, and S. Schmidt. High-resolution temperature and precipitation variability of southwest Anatolia since 1730 CE from Lake Gölcük sedimentary records. *Turkish Journal of Earth Sciences*, 30:601–610, 2021b. doi:10.3906/yer-2008-14
- Z. B. Ön, A. M. Greaves, S. Akçer-Ön, and M. S. Özeren. A Bayesian test for the 4.2 ka BP abrupt climatic change event in southeast Europe and southwest Asia using structural time series analysis of paleoclimate data. *Climatic Change*, 165:7, 2021b. doi:10.1007/s10584-021-03010-6
- S. Akçer-Ön, A. M. Greaves, S. W. Manning, Z. B. Ön, M. N. Çağatay, M. Sakınç, A. Oflaz, C. Tunoğlu, and R. Salihoğlu. Redating the formation of Lake Bafa, western Turkey: Integrative geoarchaeological methods and new environmental and dating evidence. *Geoarchaeology*, 35:659–677, 2020. doi:10.1002/gea.21791
- Z. B. Ön and M. S. Özeren. Temperature and precipitation variability in eastern Anatolia: Results from independent component analysis of Lake Van sediment data spanning the last 250 kyr BP. *Quaternary International*, 514:119–129, 2018b. doi:10.1016/j.quaint.2018.11.037
- K. K. Eriş, S. Akçer-Ön, M. N. Çağatay, U. B. Ülgen, Z. B. Ön, Z. Gürocak, T. N. Arslan, D. B. Akkoca, E. Damcı, M. İnceöz, and Ö. Okan. Late Pleistocene to Holocene paleoenvironmental evolution of Lake Hazar, Eastern Anatolia, Turkey. *Quaternary International*, 486:4–16, 2017. doi:10.1016/j.quaint.2017.09.027
- Z. B. Ön, S. Akçer-Ön, M. S. Özeren, K. K. Eriş, A. M. Greaves, and M. N. Çağatay. Climate proxies for the last 17.3 ka from Lake Hazar (Eastern Anatolia), extracted by independent component analysis of μ-XRF data. *Quaternary International*, 486:17–28, 2017b. doi:10.1016/j.quaint.2017.08.066
- Z. B. Ön, M. S. Özeren, S. Akçer-Ön, and M. N. Çağatay. Spectral features of 250 kyr BP long Lake Van sediments: Milankovitch cycles and their harmonics. *Geological Bulletin of Turkey*, 60(4):471–488, 2017c. doi:10.25288/tjb.360604
- J. Kaiser, Z. B. Ön, H. W. Arz, and S. Akçer-Ön. Sedimentary lipid biomarkers in the magnesium rich and highly alkaline Lake Salda (south-western Anatolia). *Journal of Limnology*, 75:581–596, 2016. doi:10.4081/jlimnol.2016.1337

Book Chapters

- Z. B. Ön and S. Akçer-Ön. Bayesian approaches for imputing missing data in environmental studies: Hierarchical models and Kalman filter techniques. In L. Giuliano, editor, *CIESM Monograph 53*, pages 135–144. CIESM Publisher, 2026. In: CIESM Monograph 53, 162 p
- G. D'Auria, M. El Bour, P. Furla, D. Giovannelli, L. Giuliano, M. Jebber, P. Moschella, Z. B. Ön, L. Pinheiro, K. Rassoulzadegan, M. Reyes Prieto, L. Richardson, T. Schmitt, J. Tamames, V. Tassin Campanella, A. Vitale Brovarone, M. Yakimov, M. Yücel, and N. Zaaboub. An ocean of gradients – towards a 3D mapping and visualization of high-resolution marine data – an overview. In L. Giuliano and A. Rodriguez y Baena, editors, *CIESM Monograph 53*, pages 5–28. CIESM Publisher, Paris and Monaco, 2026. In: CIESM Monograph 53, 162 p
- S. Akçer-Ön, Z. B. Ön, and A. M. Greaves. Paleoiklim ve paleoçevre çalışmalarında duraylı oksijen ve karbon izotop analizlerinin kullanımı. In Y. S. Erdal, G. D. Arıkan, and Ç. Atakuman, editors, *Arkeolojide Çok Disiplinli Yaklaşımlar*, pages 237–253. Ege Yayınları, 2024a. ISBN 9786256212077. (in Turkish)

Comments & Replies

- Z. B. Ön, A. M. Greaves, S. Akçer-Ön, and M. S. Özeren. Comment on H. Weiss' review of "Is there evidence for a 4.2 ka B.P. event in the northern North Atlantic region?". *Climate of the Past Discussions*, 2019a. doi:10.5194/cp-2018-162-SC2

Popular Articles

- Z. B. Ön. Karotlarda saklı iklim hikâyeleri (in Turkish). *Felsefe Bülteni*, 2, 2026a. URL https://felsefebulteni.com/bulten/14. Popular article
- Z. B. Ön. How X-ray fluorescence methods reveal past climate changes, 2017b. URL https://drive.google.com/file/d/1vvj4ypYc1aZEZAKmtDzfBoPgdTfBjM58/view?usp=drive_link. Popular article

Abstracts / Conference Presentations

- C. B. Bora, M. E. Ateş, and Z. B. Ön. Lisansüstü felsefe tezlerinin son 50 yılı (1974-2024): Tematik bir analiz. In *II. Sosyal Bilimler Lisansüstü Öğrenci Sempozyumu*, Erzurum, Türkiye, 2025
- H. Çaldırak, Z. B. Ön, M. S. Akkiraz, M. N. Çağatay, B. Ghaleb, S. Wulf, K. K. Eriş, D. Acar, J. Kaiser, and S. Akçer-Ön. Chronology and environmental changes from a sediment core spanning the last 487 ka from Lake Acıgöl (SW Anatolia). In *EGU General Assembly*, Vienna, Austria, 2025b
- H. Okur, M. K. Erturaç, M. Çelen, E. Şahiner, Z. B. Ön, S. Akçer-Ön, N. Köse, H. T. Güner, N. Karlıoğlu-Kılıç, and M. S. Öncel. Multiproxy evaluation of the paleohydrology of the Sakarya River during the last milennium, NW Anatolia. In *EGU General Assembly*, Vienna, Austria, 2025
- S. Akçer-Ön, Ö. Sert, Z. B. Ön, and N. Köse. Resilience of istanbul: Towards a multiproxy analysis depending on documentary sources, dendrochronology data and Küçükçekmece lagoon sediment records in the last millennium. In *76th Geological Congress of Turkey*, Ankara, Turkey, 2024b. April 15–19
- Z. B. Ön, M. E. Ateş, and J. Kaiser. Can we define the onset of the Anthropocene through statistics? In *76th Geological Congress of Turkey*, Ankara, Turkey, 2024. April 15–19
- H. Çaldırak, Z. B. Ön, and S. Akçer-Ön. Reconstructing the last 480,000 years: The age-depth model of a 128 m long core section from Lake Acıgöl, western Anatolia. In *76th Geological Congress of Turkey*, Ankara, Turkey, 2024. Poster; April 15–19
- Z. B. Ön. Critical thinking and uncertainty quantification in Quaternary research: Insights from the 4.2 ka BP event and Lake Bafa's closure date. In *7th Research Network "Socio-ecological approaches to the history and archaeology of Anatolia"*, İstanbul, Turkey, 2023b. November 20–21
- Z. B. Ön, N. Macdonald, S. Akçer-Ön, and A. M. Greaves. A novel Bayesian multilevel regression approach to the reconstruction of an eastern Mediterranean temperature record for the last 10,000 years. In *24th Paleontology and Stratigraphy Workshop*, Mardin, Turkey, 2023a. October 6–8
- Z. B. Ön. Refining the dates of Holocene environmental changes in Lake Bafa: A hierarchical Bayesian analysis of change points in proxy data. In *Sedimentology Working Group*, İstanbul, Turkey, 2023c. September 7–10

- H. Okur, M. K. Erturaç, M. Çelen, E. Şahiner, Z. B. Ön, S. Akçer-Ön, and N. Köse. The last 600-year hydrological history and extreme events of NW Anatolia. In *75th Geological Congress of Turkey*, Ankara, Turkey, 2023. April 10–14
- I. B. Danladi, S. Akçer-Ön, T. Litt, Z. B. Ön, and L. Wacker. Determination of late Holocene climate changes from the sedimentary records of the Isparta Gölcük Lake (SW Anatolia). In *75th Geological Congress of Turkey*, Ankara, Turkey, 2023a. April 10–14
- Z. B. Ön. Bayesian change point analysis for evaluation of the 4.2 ka BP climate event in southeast Europe and southwest Asia. In *75th Geological Congress of Turkey*, Ankara, Turkey, 2023a. April 10–14
- H. Okur, M. K. Erturaç, M. Koçal-Çelen, E. Şahiner, A. Gürbüz, Ö. Makaroğlu, S. Akçer-Ön, N. Karlıoğlu-Kılıç, B. A. Gümüş, H. T. Güner, Z. B. Ön, E. Bulut, F. Küçük, B. A. Gümüş, B. B. Babadağ, D. Mouralis, N. Köse, and M. S. Öncel. Investigation of past hydrological extremes (HEX) using river flood plain sediments: Sakarya River, NW Türkiye. In *6th Asia-Pacific Conference on Luminescence and Electron Spin Resonance Dating (APLED)*, Ankara, Turkey, 2022. September 26–28
- Z. B. Ön, N. Macdonald, S. Akçer-Ön, and A. M. Greaves. Reconstruction of eastern Mediterranean Holocene temperature through Bayesian hierarchical regression of temperature data. In *The Future of Ancient Forests*, Liverpool, UK, 2022. June 20
- S. Akçer-Ön, A. M. Greaves, S. W. Manning, Z. B. Ön, M. N. Çağatay, M. Sakınç, A. Oflaz, C. Tunoğlu, and R. Salihoğlu. When did the Gulf of Latmos isolate from the Aegean Sea? In *The Future of Ancient Forests*, Liverpool, UK, 2022b. June 20
- S. Akçer-Ön, N. Çağatay, M. Sakınç, Z. B. Ön, and Ö. Sert. High resolution late Holocene climatic records from Lake Salda sediments. In *The workshop of Salda and Burdur Lakes with international participation*, Burdur, Turkey, 2022a. June 5–7
- H. Okur, D. Mouralis, A. Gürbüz, M. K. Erturaç, Ö. Makaroğlu, E. Şahiner, N. Köse, N. Karlıoğlu-Kılıç, B. A. Gümüş, Z. B. Ön, and S. Akçer-Ön. Investigation of past hydrological extremes (HEX) using river flood plain sediments: Sakarya River. In *Sedimentology Working Group 2020*, Ayvalık, Turkey, 2021. September 2–5
- Z. B. Ön, M. S. Akkiraz, B. Ghaleb, S. Wulf, and S. Akçer-Ön. Chronology of a 296 m long sediment sequence spanning the last 1.35 myr from Lake Acıgöl (SW Anatolia). In *Sedimentology Working Group 2020*, Ayvalık, Turkey, 2021a. September 2–5
- I. B. Danladi, S. Akçer-Ön, T. Litt, and Z. B. Ön. Climate and environmental reconstruction through sediment records of Lake Gölcük (isparta, SW Anatolia) since the last 1600 years. In *Sedimentology Working Group 2020*, Ayvalık, Turkey, 2021a. September 2–5
- S. Akçer-Ön, N. Çağatay, M. Sakınç, Z. B. Ön, and Ö. Sert. The Little Ice Age and Medieval Climate Anomaly in İstanbul: Correlation of Küçükçekmece Lagoon sedimentary records with historical and instrumental data. In *Winds of Change: Environment and Society in Anatolia*, İstanbul, Turkey, 2021. April 29–30
- H. Okur, D. Mouralis, A. Gürbüz, M. K. Erturaç, Ö. Makaroğlu, E. Şahiner, N. Köse, N. Karlıoğlu-Kılıç, B. A. Gümüş, Z. B. Ön, and S. Akçer-Ön. Investigation of Holocene palaeo-hydrological changes using fluvial archives: A case study for Sakarya River terraces, NW Anatolia (Turkey). In *Journées des Jeunes Géomorphologues*, Paris, France, 2020. February 6–7
- Z. B. Ön, M. S. Özeren, S. Akçer-Ön, and A. M. Greaves. A Bayesian test for the southeast Europe and southwest Asia paleoclimate data used as evidences of the 4.2 ka BP event. In *4.2 ka BP in Anatolia: The Crisis Years and Aftermath*, İzmir, Turkey, 2019b. May 17–18
- Z. B. Ön. Filling the gaps in climate measurement data via Bayesian structural time series. In *GRECPIMA*, Muğla, Turkey, 2019. May 8–10
- I. B. Danladi, E. Ateş, D. Acar, S. Akçer-Ön, and Z. B. Ön. Hydroclimate changes in the Lake Gölcük (isparta, SW Anatolia) sedimentary records. In *GRECPIMA*, Muğla, Turkey, 2019b. May 8–10
- Z. B. Ön, M. S. Özeren, S. Akçer-Ön, and A. M. Greaves. A Bayesian test on the 4.2 ka BP abrupt climatic change event for the Eastern Mediterranean and Arabian Peninsula paleoclimate data using structural time series. In *SNELA*, İstanbul, Turkey, 2019c. March 13–16
- S. Akçer-Ön, A. M. Greaves, S. W. Manning, Z. B. Ön, M. N. Çağatay, M. Sakınç, A. Oflaz, C. Tunoğlu, and R. Salihoğlu. When did gulf of Latmos isolated from Aegean Sea? In *SNELA*, İstanbul, Turkey, 2019. March 13–16
- I. B. Danladi, S. Akçer-Ön, Z. B. Ön, and D. Acar. The late Holocene climate changes from Lake

Gölcük (SW Anatolia): Evidence from multi-proxy results. In *EGU General Assembly*, Vienna, Austria, 2019a. Poster; April 7–12

- Z. B. Ön and M. S. Özeren. Cold and wet glacial maximum periods in eastern Mediterranean: Independent component analysis results of 250 kyr long continuous Lake Van sediments. In *TURQUA*, İstanbul, Turkey, 2018a. May 2–5
- Z. B. Ön, M. S. Özeren, S. Akçer-Ön, and A. M. Greaves. The 4.2 ka BP event, was it really global? In *TURQUA*, İstanbul, Turkey, 2018. May 2–5
- K. K. Eriş, S. Akçer-Ön, M. N. Çağatay, and Z. B. Ön. Late Pleistocene-Holocene paleoclimate and water level changes in Lake Hazar, eastern Anatolia, Turkey. In *TURQUA*, İstanbul, Turkey, 2018. May 2–5
- T. Korkmaz, Z. B. Ön, and S. Akçer-Ön. Preliminary results on tephrochronological record of Lake Acıgöl. In *TURQUA*, İstanbul, Turkey, 2018. Poster; May 2–5
- S. Akçer-Ön, A. M. Greaves, Z. B. Ön, S. W. Manning, M. N. Çağatay, M. Sakınç, and C. Tunoğlu. When did Latmos gulf close? Geology versus Archaeology. In *International Open Workshop – Socio-Environmental Dynamics over the Last 12,000 Years: The Creation of Landscapes V*, Kiel, Germany, 2017a. March 20–24
- Z. B. Ön. Croll-Milankovitch hypothesis: Its past, problems and applications. In *70th Geological Congress of Turkey*, Ankara, Turkey, 2017a. April 10–14
- Z. B. Ön, S. Akçer-Ön, M. S. Özeren, K. K. Eriş, A. M. Greaves, and M. N. Çağatay. Revealing the precipitation and temperature records of Lake Hazar (Eastern Anatolia) from μ-XRF data, for the last 17.8 ka via independent component analysis. In *International Open Workshop – Socio-Environmental Dynamics over the Last 12,000 Years: The Creation of Landscapes V*, Kiel, Germany, 2017a. Poster; March 20–24
- S. Akçer-Ön, Z. B. Ön, M. S. Akkiraz, C. Helvacı, M. N. Çağatay, M. Sakınç, K. K. Eriş, D. Acar, İ. Gündoğan, S. Wulf, Ü. Sancar, J. García-Veigas, Y. Büyükmeriç, I. B. Danladi, R. Salihoğlu, Ö. L. Şen, M. Özdoğan, and O. Appelt. High resolution climatic & environmental records of western Anatolia for the last 320 ka: Acıgöl lake sediments. In *70th Geological Congress of Turkey*, Ankara, Turkey, 2017b. Poster; April 10–14
- S. Akçer-Ön, Z. B. Ön, M. N. Çağatay, K. K. Eriş, M. Sakınç, S. Wulf, C. Helvacı, D. Acar, İ. Gündoğan, M. S. Akkiraz, Ü. Sancar, J. García-Veigas, I. B. Danladi, R. Salihoğlu, Y. Büyükmeriç, O. Appelt, Ö. L. Şen, and M. Özdoğan. High resolution middle-late Holocene climatic records from Lake Salda (W. Anatolia) sediments. In *70th Geological Congress of Turkey*, Ankara, Turkey, 2017c. Poster; April 10–14
- S. Akçer-Ön, M. N. Çağatay, M. Sakınç, Z. B. Ön, D. Acar, I. B. Danladi, and C. Helvacı. High resolution climatic records of Western Anatolia for the last 700 ka: Acıgöl Lake sediments. In *MedCLIVAR*, Athens, Greece, 2016. September 26–30
- Z. B. Ön, S. Akçer-Ön, M. S. Özeren, and K. K. Eriş. Independent component analysis of μ-XRF time-series data: an example from Lake Hazar. In *TURQUA*, İstanbul, Turkey, 2016. May 8–11
- K. K. Eriş, S. Akçer-Ön, Z. B. Ön, and M. N. Çağatay. Late Pleistocene to Holocene paleoenvironmental evolution of Lake Hazar, Eastern Anatolia, Turkey; evidences from multi-proxy records. In *TURQUA*, İstanbul, Turkey, 2016. May 8–11
- I. B. Danladi, R. Salihoğlu, Z. B. Ön, and S. Akçer-Ön. High-resolution multi proxy evidences of climate changes during the past 1400 years from Lake Salda sediment records (SW Turkey). In *MedCLIVAR*, Athens, Greece, 2016. Poster; September 26–30
- S. Akçer-Ön, A. M. Greaves, M. N. Çağatay, M. Sakınç, Z. B. Ön, and C. Tunoğlu. When did Latmos gulf close? Geology versus archaeology. In *68th Geological Congress of Turkey*, Ankara, Turkey, 2015. April 6–10
- D. Türkmen, Ş. Şen, S. Akçer-Ön, Z. B. Ön, and T. Önal-Süzek. Mobile application based geological core database system. In *Genç Yerbilimciler Kongresi*, İzmir, Turkey, 2015. Poster; June 5–7
- T. L. Erel, B. Batır, and Z. B. Ön. The impact of the changing coastline on the harbors in Bodrum and Datça peninsulas, and the examples of Bodrum (halikarnassos), aspat (strobilos), Datça (stadia), tekirburnu (knidos). In *8th International Symposium on Eastern Mediterranean Geology*, Muğla, Turkey, 2014. October 13–17
- Z. B. Ön, M. S. Özeren, M. N. Çağatay, and K. K. Eriş. Late Pleistocene-Holocene climatic cycles from Eastern Turkey: Lake Van and Lake Hazar sediment records. In *MedCLIVAR*, Ankara, Turkey,

2014. June 22–25

- Z. Ankut, S. Ercan, C. Eroğlu, S. Akçer-Ön, T. Korkmaz, Z. B. Ön, and D. Acar. Determination of late Holocene climatic changes at Lake Köyceğiz sedimentary records: preliminary results. In *67th Geological Congress of Turkey*, Ankara, Turkey, 2014. Poster; April 14–18
- Z. B. Ön, M. S. Özeren, M. N. Çağatay, S. Akçer-Ön, E. Damcı, and Ü. Sancar. Late Pleistocene-Holocene climatic cycles from Lake Van sediments. In *66th Geological Congress of Turkey*, Ankara, Turkey, 2013. April 1–5

Research Projects

- Z. B. Ön. Geçmişten günümüze Batı Anadolu'da iklimsel ve hidrolojik krizlerin çevresel etkilerinin çok disiplinli yöntemlerle araştırılması, 2025. Research project (2025–2028). Role: Researcher. Funder: TÜBİTAK (The Scientific and Technological Research Council of Turkey). Project Coordinator: M. Korhan Erturaç
- Z. B. Ön. Ancient and future forests, 2022. Research project (2022–2023). Role: Researcher. Funder: British Council (Ankara). Project Coordinators: Alan M. Greaves and Sena Akçer-Ön
- Z. B. Ön. Sakarya nehri güncel taşkın ovası örneğinde akarsu çökellerinin geçmiş hidrolojik ekstremler (HEX) açısından olay stratigrafisinin çoklu-vekil yaklaşımı ile belirlenme potansiyelinin araştırılması, 2021. Research project (2021–2022). Role: Researcher. Funder: TÜBİTAK (The Scientific and Technological Research Council of Turkey). Project Coordinator: M. Korhan Erturaç
- Z. B. Ön. Late Holocene climate changes and its relation with solar forcing in southwestern Anatolia from Lake Gölcük (isparta) sediment records, 2018. Research project (2018–2019). Role: Researcher. Funder: TÜBİTAK (The Scientific and Technological Research Council of Turkey). Project Coordinator: Sena Akçer-Ön
- Z. B. Ön. Palaeoclimate of southwestern Anatolia during the Quaternary period: Lake Acıgöl and Lake Salda sediment records, 2015a. Research project (2015–2017). Role: Researcher. Funder: TÜBİTAK (The Scientific and Technological Research Council of Turkey). Project Coordinator: Sena Akçer-Ön
- Z. B. Ön. High resolution climate and environment studies from sediment and water samples of Lake Salda (burdur), 2015b. Research project (2015–2017). Role: Researcher. Funder: Muğla University. Project Coordinator: Sena Akçer-Ön

Organizations

- Organizer at 8th International Symposium on Eastern Mediterranean Geology (ISEMG), October 2014, Muğla, Turkey.
- Convener at 76th Geological Congress of Türkiye (TJK), April 2024, Ankara, Turkey.

Awards & Scholarships

- Best thesis award, İTÜ, 2019.